

Esercizi

1. L'evento considerato si verifica se e solo se un numero pari di errori: la prob. sarà che

$$\sum_{k=1}^4 \binom{8}{2k} p^{2k} (1-p)^{8-2k} = 28 p^2 (1-p)^6 + 70 p^4 (1-p)^4 + 28 p^6 (1-p)^2 + p^8$$

2. $x(kT)$ viene ottenuto da $w(kT)$ tramite il filtro

$$H(f) = \frac{X(f)}{W(f)} = \frac{1}{1 - 0.9 e^{-j2\pi f T}}$$

$$h(kT) = \frac{1}{T} \cdot 0.9^k \cdot 1_0(kT)$$

$$R_w(f) = \sigma_w^2 T$$

$$R_x(f) = R_w(f) \cdot |H(f)|^2$$

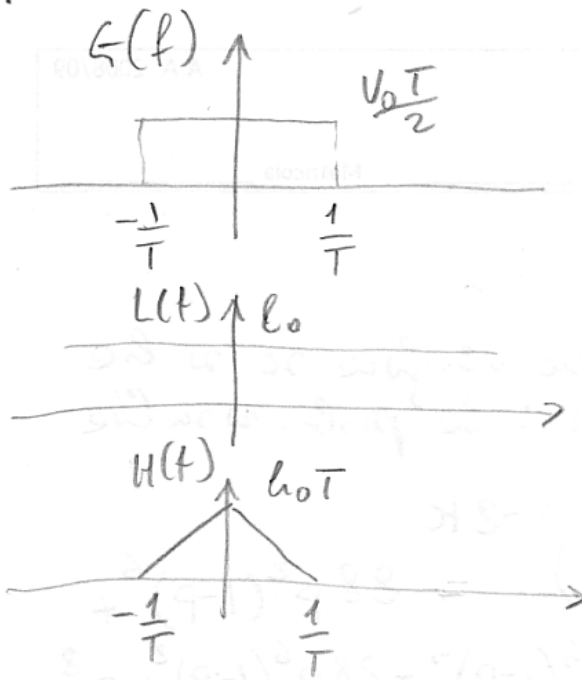
$$M_x = \int_0^{\frac{1}{T}} R_x(f) df = \int_0^{\frac{1}{T}} \sigma_w^2 T |H(f)|^2 df$$

$$\uparrow \quad \sigma_w^2 T \cdot T \cdot \sum_{k=-\infty}^{+\infty} h^2(kT) = \sigma_w^2 T^2 \sum_{k=0}^{+\infty} (0.81)^k$$

Parseval

$$= \frac{\sigma_w^2}{1 - 0.81}$$

3.



$$C(f) = \frac{V_0 T}{2} l_0 H(f)$$

$$c(t) = \frac{V_0 T}{2} l_0 h_0 \text{sinc}^2\left(\frac{t}{T}\right) \quad \text{di Nyquist}$$

$$V_0' = \frac{V_0 T}{2} l_0 h_0$$

$$\sigma_m^2 = \int_{-\infty}^{+\infty} R_0 |H(f)|^2 df = R_0 \cdot 2 \cdot l_0^2 T^2 \cdot \frac{1}{T} \cdot \frac{1}{3}$$

$$P_e = \frac{4}{3} Q\left(\frac{V_0'}{2\sigma_m}\right)$$

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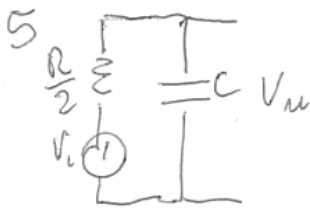
4. $y(t)$ assume valori fra $-V_0^3$ e V_0^3 con probabilità

1. Dunque $\Delta = \frac{2V_0^3}{L}$

$$M_y = E[y^2(t)] = E[x^6(t)] = \int_{-V_0}^{V_0} \frac{1}{2V_0} a^6 da$$

$$= \frac{1}{2V_0} \left. \frac{a^7}{7} \right|_{-V_0}^{V_0} = \frac{1}{2V_0} \frac{1}{7} V_0^7 \cdot 2 = \frac{V_0^6}{7}$$

$$SNR = \frac{M_y}{\Delta^2} 12 = \frac{12 V_0^6}{7} \cdot \frac{L^2}{4 V_0^6} = \frac{3}{7} L^2$$



$$V_u = \frac{1}{1 + j 2\pi f C R \frac{R}{2}} V_i$$

$$R_{v_u}(f) = 2kTR \frac{1}{2} \frac{1}{1 + \pi^2 f^2 C^2 R^2}$$

$$\sigma_{v_u}^2 = \int_{-\infty}^{+\infty} R_{v_u}(f) df = kTR \int_{-\infty}^{+\infty} \frac{1}{1 + \pi^2 f^2 C^2 R^2} df$$

$$= \left(\pi f C R \rightarrow f'; \int \frac{1}{1+f'^2} df' = \arctan f' \right) = \frac{kT}{C}$$

v_u è gaussiano e medio nullo e varianza $\sigma_{v_u}^2$

$$P[|v_u(t)| > V_0] = 2Q\left(\frac{V_0}{\sigma_{v_u}}\right)$$